HACKATHON
CHALLENGE | NEURAL WAVE 2026

From Hourly to Quarter-Hourly

Reconstruction of 15-minute electricity price profiles across multiple European markets

Challenge objective:	transform an already available daily hourly curve into a realistic quarter-hourly profile, while maintaining exact consistency with the 24 reference hourly prices.
Scope:	Time-series reconstruction | Machine Learning | electricity markets
Proponent:	Azienda Elettrica Ticinese (AET)
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1. Introduction

Azienda Elettrica Ticinese, AET, was established in 1958 with the mission of producing, transmitting and commercializing electricity. Today, AET is the leading electricity wholesaler in the Canton of Ticino and is one of the key players in the cantonal energy transition.

While the Swiss market remains its main reference market, AET also operates in several European markets and maintains bilateral and over-the counter (OTC) energy and commercial relationships with leading operators. AET is recognized as a reliable and qualified counterparty on the main European power exchanges.

Participation in power markets requires a high level of expertise and knowledge in front-office activities such as screen trading, middle-office activities such as analysis, and in back-office activities.

However, these capabilities alone are insufficient: to effectively assess market opportunities and positions, analysis and trading activities require price curves that are aligned with the granularity of traded products and operational processes.

This requirement has become particularly relevant with the transition of the European day-ahead market to quarter-hourly granularity. From 1st October 2025 onwards, prices that were previously determined mainly on an hourly basis, will be formed for 15-minute intervals. However, many forecasts, historical series and long-term curves remain available only at hourly granularity. Therefore, a method is needed to reconstruct, for each hour, a realistic quarter-hourly profile for each hour while preserving the original hourly average price.

Why the hourly price is not enough: simply replicating the hourly price over the four quarter-hours is not sufficient because, even though the average of the values is mathematically equivalent to the hourly price, it does not capture the volatile intra-hour price dynamics. . Observed data often exhibits recurring patterns, such as sawtooth profiles, which are important for position valuation, trading, and the training of 15-minute models.

2. Definitions

Definitions	Description
CET/CEST	Central European Time / Central European Summer Time. The Central European time zone, is equal to UTC+1 under standard time (CET) and UTC+2 under daylight saving time (CEST).
DST	Daylight Saving Time. A seasonal time regime in which clocks are advanced by one hour compared with standard time. In Europe it is in force from the last Sunday in March to the last Sunday in October, with the clock change taking place at 01:00 UTC.
ENTSO-E	European Network of Transmission System Operators for Electricity. It is a European association of transmission system operators (TSOs), including Swissgrid for Switzerland. Its responsibilities include operating the Transparency Platform, which publishes market and system data such as prices, load, generation, and cross-border flows.
MAE	Mean Absolute Error. An accuracy indicator equal to the mean of the absolute differences between estimated and observed values: MAE = $(1/n) \cdot \Sigma \, \lvert p_i - \hat{p_i} \rvert$. It is expressed in the same unit as the measured quantity (e.g. EUR/MWh) and, unlike other indicators, it does not amplify the weight of larger errors.
OTC	Over-the-Counter. Bilateral trading outside of regulated exchanges, concluded directly between counterparties or through brokers. Contracts may be standardized or customized, and their terms are agreed between the parties.
SDAC	Single Day-Ahead Coupling. European single day-ahead market coupling mechanism that simultaneously determines prices and cross-border electricity exchanges in participating zones simultaneously, taking available transmission capacities into account. Tt has operated at 15-minute granularity since 1 October 2025.
UTC	Coordinated Universal Time. An international reference time scale that has not subject to changes between standard time and daylight saving time. For this reason, it is commonly used for recording and exchanging time series, as it ensures a continuous and unambiguous sequence of intervals.

3. Day-ahead market context

In the day-ahead market, the spot price of electricity is determined one day before delivery. This means that electricity is actually bought and sold in advance. Market participants submit buy and sell orders for the various delivery (intervals) periods of the next day in a so-called day-ahead auction . The SDAC algorithm then uses a single market-clearing process to match supply and demand, taking into account capacity constraints between market areas.

This determines a single market price for each market area and each delivery interval. All participants in the same area therefore refer to the same price for that specific interval. The ordered sequence of these prices over the day constitutes the daily price curve.

Operational summary of the challenge

In short, the challenge involves reconstructing 15-minute intra-hour dynamics from the hourly curve that is already available , subject to the following key requirements:

Element	Essential rule
Daily input	24 hourly prices per zone, provided together.
Required output	96 15-minute prices per zone: four values for each hour.
Core constraint	For each hour, the average of the four predicted quarter-hourly prices must equal the provided hourly price.
Permitted data	Official training set, hourly curve for the evaluated day, and prior quarter-hourly history provided by the jury. No external sources.
Evaluation	MAE on 15-minute prices, with equal weight for the four zones, together with a qualitative assessment of robustness, clarity and reproducibility.

4. Information available in the challenge

The day-ahead auction jointly determines the prices for each delivery interval for the next day. Participants will therefore receive **all 24 hourly prices** for each zone and evaluation date **together**, so they do not need to forecast subsequent hourly prices. Participants may process each hour separately or the entire curve, using the method they consider most appropriate. During the training process, only the data included in the official data package provided by organizers may be used.

5. Challenge objective

Participants must develop a model that, for each market area and day, transforms a curve composed of 24 hourly prices into 96 15-minute prices, i.e. four values for each hour.

Daily input	Required transformation	Daily output
24 hourly prices	Reconstruction of the four quarter-hours with unchanged average	96 15-minute prices

To keep the dataset uniform, only days representable by 24 hours and 96 15-minute intervals will be included. Days affected by the daylight-saving-time transition, which result in 23 or 25 hours, will be excluded from the official dataset.

Core constraint: for each hour, the average of the four reconstructed quarter-hourly prices must equal the reference hourly price. The model learns the shape within the hour, not a new hourly level.

Illustrative example

The following chart shows an actual day for Zone 0. The observed 15-minute price line shows marked intra-hour dynamics; the reference hourly curve is represented as a flat baseline, replicated over the four quarter-hours. The task is to reconstruct the quarter-hourly profile without changing the average of each hour.



Figure 1. Zone 0, 19 June 2026 — observed 15-minute prices and hourly price replicated over the four quarter-hours.

Data source: ENTSO-E Transparency Platform; AET processing.

6. Target definition

The official target comprises the four observed 15-minute prices for each hour. The model must produce four values, the average of which equals the provided hourly price. There is no specific decomposition or parameterization prescribed: teams are free to choose how to represent and estimate the intra-hour profile.

Constraint for each hour h: (1/4) * sum[q=1..4] P_hat(h,q) = H(h)

In the formula, H(h) denotes the provided hourly price and P_hat(h,q) the predicted price for quarter-hour q. The constraint defines the required output, but not the method used to obtain it.

7. Data availability

The official training set comprises a single historical file containing 15-minute day-ahead prices for the following four zones, identified as Zone 0, Zone 1, Zone 2 and Zone 3. There are 363 valid days available for each zone, equating to a total of 139,392 quarter-hourly prices. The hourly training curve can be calculated as the arithmetic mean of the four prices within the same hour.

Data source and processing: the day-ahead prices used for the challenge originate from the ENTSO-E Transparency Platform. AET subsequently selected, processed and reorganized the data for the purposes of the challenge. In particular, market areas were distributed to participants using anonymized identifiers (ZONE_0, ZONE_1, ZONE_2, ZONE_3). References to ENTSO-E exclusively identify the source of publication of the data and do not imply sponsorship, approval or endorsement of the challenge by ENTSO-E.

Time alignment: daily curves group together intervals belonging to the same delivery date. Source timestamps are normalized to CET/CEST market time and observations are not solely aggregated on the basis of the UTC date.

Dataset specifications

Element	Specification
Official training set	The period covers delivery dates from 1 October 2025 to 30 September 2026. Excluding the two days affected by the daylight-saving-time transition (DST), the official training set includes 363 valid delivery dates. For each included zone, one daily curve will be available for each of these dates, always consisting of 24 hours and 96 15-minute intervals.
Period excluded from participant use	From 1st to 18th October 2026. Before the freeze, participants may not consult or use data from these dates, even if public. After the freeze, the jury may provide the frozen pipeline with 15-minute prices only for dates preceding the evaluated day, for the calculation of lagged variables. This access does not authorize retraining, calibration or changes to the solution. The period may also be used by the organizers for stability and integrity checks.
Secret evaluation	The secret evaluation will certainly include the delivery date of 19th October 2026 and may include additional delivery dates not communicated in advance. The corresponding quarter-hourly targets will not be available to participants at the time the solution is frozen.
Unit of measure	EUR/MWh.
Target granularity	Four 15-minute prices for each hour of the daily curve.
Reference hourly price	Average of the four 15-minute prices within the hour; provided in the test set.

The training set contains only the file train_prices_15min.csv (delivery_date, market_id, hour_index, quarter_index, price_eur_mwh). The market_id field takes the values ZONE_0, ZONE_1, ZONE_2 and ZONE_3. For each evaluated date, the jury will provide the pipeline with test_hourly.csv (delivery_date, market_id, hour_index, hourly_price_eur_mwh) and history_15min.csv, with the same schema as the training file but limited to earlier dates. The quarter-hourly prices for the evaluated date will remain hidden.

Dataset design choice: a single quarter-hourly file is sufficient for training, from which the hourly average can be calculated. During evaluation, the day to be reconstructed is available only as an hourly curve; the quarter-hourly history contains earlier dates. Only inputs derived from the provided data are allowed.

Data design principle

Load, generation, and weather forecasts are not provided to the participants. The reason for this is that the solution must also be applicable to various types of forward curve and to long-term horizons for which thesevariables/forecasts are simply not available. Teams are free to choose variables derived solely from the permitted data and the 15-min the reconstruction method.

Use of historical data during inference.

During development, participants may only use the official training set for training, calibration and hyperparameter selection . However, as part of its inference logic, the solution may include the use of lagged variables or other features derived from quarter-hourly history prior to the date to be reconstructed. During evaluation, history_15min.csv may therefore also include observations after the official training period and before the evaluated delivery date. These observations can only be used as inputs to the frozen pipeline according to rules established before the freezing. They cannot be used to retrain the model, recalibrate parameters, modify hyperparameters or adapt the methodology.

For instance, when evaluating the delivery date of 19 October 2026, the pipeline may use the quarter-hourly prices of earlier dates, including those after 30 September 2026, as lags, provided this is accounted for in the submitted pipeline and does not require updating the model.

8. Input rules

Availability rule

For each zone and evaluated date, the pipeline will receive the 24 hourly prices for the day and a history of 15-minute prices from earlier dates. Teams may use the entire hourly curve and construct lagged variables from the history. Historical data provided after the freeze, even if more recent than the training set, may feed only inference rules already fixed in the submitted code: they may not be used to retrain the model, calibrate parameters or modify the rules. External sources and quarter-hourly prices for the day being reconstructed are prohibited.

Pretrained models and use of external code

Pretrained models, external weights, transfer learning, embeddings or components trained on data not included in the official package are not allowed. Standard libraries, statistical methods, traditional Machine Learning models and code written by participants are allowed, provided that training, validation, calibration and hyperparameter selection depend exclusively on the official challenge training set.

Simple, robust, interpretable and easily verifiable solutions will be viewed positively by the jury. Model complexity is not in itself a merit and must be justified by the benefits obtained.

9. Pipeline output and execution

During the evaluation stage, participants will not directly compute predictions themselves for the secret evaluation day(s). The final evaluation will be carried out by the jury by using the code, model and artifacts submitted by the teams.

The solution must therefore include a complete, documented and easily executable inference pipeline that, for each evaluated date, receives test_hourly.csv and history_15min.csv provided by the organizers as input and automatically produces the required quarter-hourly predictions. Execution must be possible without code modifications, without network access and without undocumented manual intervention. Teams must clearly specify the command, script or entry point required to start inference.

The pipeline must produce a file containing one price prediction for each delivery_date, market_id, hour_index and quarter_index.

Field	Description	Constraint
delivery_date	Delivery date (YYYY-MM-DD)	Present in the test set
market_id	Zone: ZONE_0, ZONE_1, ZONE_2 or ZONE_3	Present in the test set
hour_index	Delivery hour from 1 to 24	Chronological order
quarter_index	Quarter-hour from 1 to 4	Four rows per hour
predicted_price_eur_mwh	Estimated quarter-hourly price	Average of the four values equal to the hourly price

Before scoring, the organizers will verify that for each delivery_date, market_id and hour_index there are exactly four finite numerical prices, one for each quarter_index from 1 to 4. The absolute difference between their average and hourly_price_eur_mwh may not exceed 0.0001 EUR/MWh. Incomplete, duplicated or out-of-tolerance outputs will be considered non-compliant and will not proceed to official scoring.

No automatic correction of invalid outputs is provided. The flat baseline remains a comparison method that teams may implement, not a fallback value inserted by the organizers.

The ability of the solution to be reproduced and executed quickly by the jury is an integral part of project evaluation.

10. Evaluation

Evaluation framework

Level	What is checked	Effect
Eligibility	Complete and numeric output; four quarter-hours per hour; compliance with the hourly-average constraint; exclusive use of permitted inputs.	A violation results in non-compliance or exclusion from official scoring.
Quantitative evaluation	MAE of quarter-hourly prices, calculated by day and zone and then averaged with equal weight across the four zones.	Determines the quantitative ranking: a lower value is better.
Qualitative evaluation	Justified simplicity, robustness, interpretability, methodological clarity and pipeline reproducibility.	Contributes to the jury's final evaluation together with the quantitative score.

Quantitative evaluation

The quantitative metric is the Mean Absolute Error (MAE), i.e. the mean absolute difference between estimated and observed 15-minute prices, expressed in EUR/MWh. The MAE is first calculated for each day and zone, then averaged across days for each zone. The official score is the simple average of the MAEs of the four zones: each has a 25% weight and a lower value is better.

MAE_day = mean over 96 quarter-hours |P_hat(d,h,q) - P_obs(d,h,q)|

MAE_zone = mean over days MAE_day

Score = mean over zones MAE_zone

The quantitative score will be calculated exclusively by the organizers by running the submitted solution on the secret evaluation dataset.

Diagnostics published together with the score:

- error by zone and by quarter-hour position within the hour;
- error distribution and analysis of the highest-error cases;
- compliance with the constraint that keeps the hourly average unchanged;
- comparison with the flat baseline, which replicates the hourly price over the four quarter-hours.

Qualitative evaluation

The quantitative ranking based on MAE is one component of the final evaluation and does not alone determine the winner. The jury will also qualitatively assess the simplicity of the approach, its robustness, its interpretability,the clarity of the methodology and the reproducibility of the pipeline. The extent to which any introduced complexity is actually justified by the results obtained will also be considered.

The qualitative evaluation will be performed on the submitted solution, code and methodology document . A simpler approach does not automatically receive a better evaluation, but unnecessary or difficult-to-verify complexity may be penalized relative to a solution with comparable predictive quality that is more transparent.

11. Evaluation protocol and reproducibility

The rules distinguish between the development stage prior to the freezing of the pipeline and its subsequent use during testing. All model choices must be reproducible from the official training set. The term “frozen pipeline” refers to the entire submitted solution, comprising the model, parameters and hyperparameters, feature engineering, rules for utilising historical data, post-processing and any additional logic that could influence predictions. After the freeze, these components may not be modified or adapted based on evaluation data.

Phase	Allowed or required	Not allowed
Development before the freeze	Training from 1st October 2025 to 30th September 2026, excluding the two DST days. From these data: validation, model and hyperparameter selection, training and calibration.	Consulting or using, even indirectly, prices from 1–18 October 2026, external data, online archives or pretrained weights.
Submission and freeze	Code, configuration, dependencies, seeds and regenerable artifacts. Separate entry points for training and inference; pipeline executable without undocumented manual intervention.	Modifying the model, parameters, features, post-processing or other rules after submission based on information relating to the test set.
Inference after the freeze	Use the 24 hourly prices for the evaluated date and history_15min.csv provided by the jury, including dates after the training period but always before the evaluated date. The history may feed rules already fixed in the code.	Using quarter-hourly prices for the evaluated date, retraining, recalibrating, selecting new hyperparameters, or performing external lookups and scraping.

The secret evaluation will certainly include the delivery date of 19 October 2026, and may include additional delivery dates that are not communicated in advance. For each date, the jury will only provide the pipeline with the hourly prices for that day and quarter-hourly history limited to earlier dates.

The confirmed inclusion of 19 October 2026 does not imply that the solution should be optimized specifically for that date: robustness may also be assessed on additional secret delivery dates not communicated in advance.

The organizers may use the period from 1 to 18 October 2026 for stability and integrity checks. An anomalous deviation from the results on the secret dates will not by itself constitute evidence of a violation, but may prompt a review of the submitted code and artifacts.

The official baseline replicates the hourly price over the four quarter-hours. The jury may retrain the solution from scratch using only the official training set and run inference without network access. A violation of the input rules will result in exclusion from official scoring.

12. Expected submission

Each team must submit a self-contained solution including:

- complete code required for training and inference;
- clearly identified entry points to retrain the model from scratch and to run predictions on the test set provided by the organizers;
- dependency files and environment configuration required for execution;
- any artefacts produced during training and required for inference, provided that they can be regenerated by the submitted training pipeline;
- a methodology note describing the approach used and justifying the main design choices, including features, the model, validation and post-processing;

This documentation should enable the jury to quickly understand the purpose of the model, the rationale behind the design choices, how to retrain it, and how to run it on new data. Preference will be given to solutions in which the introduced complexity is proportionate to the benefit obtained and in which training and inference can be executed reliably with a minimum number of manual steps.

13. Essential FAQ

Does the challenge also involve forecasting hourly prices?

No, the hourly curve is an input that is already available. The task is to reconstruct its 15-minute shape exclusively.

Does the model have to process each hour separately?

No, all 24 hourly prices for the same day are provided together, so the team can model individual hours or the entire curve. They may also use the quarter-hourly history made available to the pipeline by the jury, but this is always limited to earlier dates and does not include the 15-minute prices for the day (being predicted) in question.

Is it possible to train a single model for all zones?

Yes., both a single model for all zones, using “market_id” to distinguish them, and separate models are permitted. The average of the MAEs of the four zones gives each zone the same weight in the official score.

Are additional data such as demand, generation or weather provided?

No, the official package only includes the 15-minute training prices. During the evaluation, the jury will provide the hourly prices for the days to be reconstructed, as well as the prior quarter-hourly history. Load, generation and weather data are not provided, and variables may not be obtained from external sources.

Are pretrained models allowed?

No. To ensure comparability and reproducibility, particpants must exclusively use the official challenge training set and code that has not been trained on external series for training, validation, calibration and hyperparameter selection.